

Aetherian Dynamics Technical Brief

Evaluation corpus edition. All facts in this brief are fictional and self-contained.

1. Company overview

Aetherian Dynamics is a fictional aerospace materials company founded in 1998 in Althaven, the capital of the Republic of Sereleia. The company employs 1,240 people across three sites and reported revenue of 312 million sereleian crowns in the last fiscal year. Its flagship product is the Skyweave composite, a carbon-basalt laminate used in high altitude airship hulls.

2. The Skyweave composite

Skyweave is produced in sheets of 2.4 by 1.2 metres with a nominal thickness of 3.2 millimetres. The laminate has a tensile strength of 1,850 megapascals and a density of 1.42 grams per cubic centimetre. Sheets are cured at 185 degrees Celsius for 90 minutes in an autoclave. The material is certified for continuous service between minus 60 and plus 140 degrees Celsius.

3. Manufacturing sites

The Althaven plant produces raw laminate and operates two autoclaves with a combined capacity of 60 sheets per day. The Ostera plant, located next to the geothermal fields, performs the curing step using geothermal steam and reduces the energy cost of curing by 38 percent compared with the Althaven plant. The Sydra site is a research laboratory with 85 staff and no production line.

4. Quality control

Every sheet receives a serial number that encodes the plant, the autoclave and the curing batch. A sheet is rejected when its measured tensile strength falls below 1,700 megapascals or when ultrasonic inspection finds a void larger than 0.8 millimetres. The rejection rate in the last fiscal year was 2.1 percent.

5. Customers and delivery

The largest customer is the Sereleian Airship Authority, which buys about 40 percent of the annual output. International customers receive shipments through the port of Vestera; the standard lead time is 6 weeks for orders under 500 sheets and 10 weeks for larger orders.

6. Safety notes

Skyweave dust is an irritant. Cutting must be performed with wet saws or in enclosures with extraction, and operators wear P3 respirators. The safety data sheet lists a workplace exposure limit of 1 milligram per cubic metre averaged over eight hours.

7. Research roadmap

The Sydra laboratory is developing Skyweave II, which replaces the basalt layer with a recycled glass layer. Early prototypes reach 1,600 megapascals, which is below the current product, and the team expects to close the gap by the end of the roadmap period. A second project explores self-healing resin systems that seal micro cracks when heated above 70 degrees Celsius.

8. Contact and governance

The company is governed by a board of nine members, five of whom are elected by employees. Technical enquiries are handled by the materials engineering office in Althaven. The company

does not publish a public price list; quotations are issued per order.